

Variance and Central Limit Theorem for the Log Coprime Density of Random LCMs

Muhamad Wakid 

Abstract

Let $X_1, X_2, \dots$ be independent and identically distributed random variables uniform on $\{2, 3, \dots, M\}$, let $L_k = \text{lcm}(X_1, \dots, X_k)$, and let

$$W_k(M) := \ln(\varphi(L_k)/L_k) - \mathbb{E} \ln(\varphi(L_k)/L_k)$$

denote the centred log coprime density, with $s_k^2 := \text{Var } W_k(M)$. A companion paper [12] established an exact subset identity for $\rho_k(M) = \mathbb{E}[\varphi(L_k)/L_k]$, a sharp asymptotic equivalence $\rho_k(M) \rightarrow P_M$ at fixed M , and the joint-regime convergence $\ln M \cdot (\rho_{\lfloor cM \rfloor}(M) - P_M)/P_M \rightarrow A(c)$ with $A(c) := \sum_{j \geq 1} e^{-cj} \ln(1 + 1/j)$. We study second-order fluctuations in the same joint regime $k = \lfloor cM \rfloor$, $M \rightarrow \infty$, with fixed $c > 0$. We prove two results. First, the variance asymptotic

$$M \ln M \cdot s_k^2 \longrightarrow B(c) := \frac{1}{e^c - 1} - \frac{1}{e^{2c} - 1},$$

via a diagonal / off-diagonal decomposition where the off-diagonal term is shown to be negligible by an exact covariance identity valid for prime pairs (p, q) with $pq > M$. Second, a central limit theorem: $W_k(M)/s_k \xrightarrow{d} \mathcal{N}(0, 1)$ as $M \rightarrow \infty$ in the joint regime. The CLT proof uses the Hall–Heyde martingale central limit theorem applied to the Doob filtration $\mathcal{F}_i = \sigma(X_1, \dots, X_i)$. Two ingredients enter. A uniform increment bound $\max_i |D_i| = O((\log M)^3/M) = o(s_k)$ on a super-polynomially likely event combines a coupon-collector tail estimate with a sieve-geometric observation $((M/(\log M)^2)^2 > M$ for M large), either fact alone being insufficient. Conditional-variance concentration is obtained from explicit moment bounds for the hitting-time variables $T_p := \min\{i : p \mid X_i\}$: a per-prime variance bound $\text{Var}(R_{pp}^{(k)}) \leq 3q_p^k/\pi_p^2$ and a cross-prime covariance bound $|\text{Cov}(R_{pp}^{(k)}, R_{qq}^{(k)})| \leq C_0(c)kq_p^kq_q^k$ valid when $pq > M$ and $\pi_p + \pi_q \leq 1/3$, with an explicit constant $C_0(c)$ depending only on c , together with an L^2 Minkowski summation that controls the covariances among all distinct prime-pair terms.

Keywords: Random LCM, Euler totient, coprime density, central limit theorem, martingale central limit theorem, Doob decomposition, Hall–Heyde, probabilistic number theory.

MSC 2020: 60F05, 11N37, 11K65, 60G42.

1 Introduction

The arithmetic structure of the least common multiple of a random set of integers has been studied extensively in the regime where the support size n grows while the number of samples k stays fixed or bounded; see Cilleruelo, Rué, Šarka, and Zumalacárregui [4]; Hilberdink and Tóth [6]; Alsmeyer, Kabluchko, and Marynych [1]; Bostan, Marynych, and Raschel [2]; Buraczewski, Iksanov, and Marynych [3]; Iksanov, Marynych, and Raschel [7]; Kabluchko,

Marynych, and Raschel [8]; and Sanna [10]. These works establish limit theorems for $\log \text{lcm}$ and for multiplicative functions of the LCM in the fixed- k , $n \rightarrow \infty$ regime.

A complementary regime, in which the support $\{2, \dots, M\}$ is fixed and the sample size $k \rightarrow \infty$, was introduced in the companion paper [12]. There the natural object of study is not the LCM itself but the coprime density

$$\frac{\varphi(L_k)}{L_k} = \prod_{p|L_k} \left(1 - \frac{1}{p}\right),$$

whose large- k behaviour is governed by exactly which primes $p \leq M$ appear as divisors of $L_k = \text{lcm}(X_1, \dots, X_k)$. The paper [12] established, writing $\rho_k(M) := \mathbb{E}[\varphi(L_k)/L_k]$, $P_M := \prod_{p \leq M} (1 - 1/p)$, $m_p := \lfloor M/p \rfloor$, $\pi_p := m_p/(M-1)$, $q_p := 1 - \pi_p$:

- *Exact identity:*

$$\rho_k(M) = P_M \sum_{T \subseteq \mathcal{P}_M} \beta_T(M)^k \prod_{p \in T} (p-1)^{-1},$$

where $\beta_T(M) = \mathbb{P}(\gcd(X, \prod_{p \in T} p) = 1)$ for a single uniform draw X .

- *Fixed- M limit with sharp rate:* $\rho_k(M) \rightarrow P_M$ as $k \rightarrow \infty$, with an explicit geometric rate of convergence controlled by $(q^{**}(M)/q^*(M))^k$.
- *Joint-regime first-order asymptotic:* in the regime $k = \lfloor cM \rfloor$, $M \rightarrow \infty$, fixed $c > 0$,

$$\ln M \cdot \frac{\rho_k(M) - P_M}{P_M} \rightarrow A(c) := \sum_{j \geq 1} e^{-cj} \ln(1 + 1/j).$$

Dependence on the companion paper. The proofs below are self-contained apart from three explicitly identified inputs from [12]: the subset-density bounds collected in Lemma 3, the level-count estimate quoted in Lemma 4, and the elementary disjointness observation quoted in Proposition 5. The exact identity and first-order asymptotic for $\rho_k(M)$ are recalled only to situate the present second-order results and are not used in the proofs of Theorems 1 and 2. The disjointness identity needed later is also rederived directly in Lemma 7.

In this paper we study the second-order behaviour of the centred log coprime density

$$W_k(M) := \ln(\varphi(L_k)/L_k) - \mathbb{E} \ln(\varphi(L_k)/L_k),$$

in the same joint regime $k = \lfloor cM \rfloor$, $M \rightarrow \infty$, with fixed $c > 0$. We prove two results.

Theorem 1 (Variance asymptotic). *For each fixed $c > 0$, in the joint regime $k = \lfloor cM \rfloor$, $M \rightarrow \infty$,*

$$M \ln M \cdot s_k^2 \rightarrow B(c) := \sum_{j \geq 1} e^{-cj} (1 - e^{-cj}) = \frac{1}{e^c - 1} - \frac{1}{e^{2c} - 1}, \quad (1)$$

where $s_k^2 = \text{Var } W_k(M)$.

Theorem 2 (Central limit theorem). *For each fixed $c > 0$, in the joint regime $k = \lfloor cM \rfloor$, $M \rightarrow \infty$,*

$$\frac{W_k(M)}{s_k} \xrightarrow{d} \mathcal{N}(0, 1).$$

Consequently,

$$\sqrt{M \ln M} \cdot \frac{W_k(M)}{\sqrt{B(c)}} \xrightarrow{d} \mathcal{N}(0, 1), \quad \sqrt{M \ln M} \cdot \frac{\varphi(L_k)/L_k - \rho_k(M)}{P_M \sqrt{B(c)}} \xrightarrow{d} \mathcal{N}(0, 1).$$

The first consequence follows from Theorem 1 and Slutsky's theorem; the second follows from the linear-scale transfer Lemma 18 below.

1.1 Method and outline

Theorem 1 is proved by a direct diagonal / off-diagonal decomposition. Writing $X_p = \mathbf{1}\{p \mid L_k\}$ so that $\ln(\varphi(L_k)/L_k) = \sum_{p \leq M} X_p \ln(1 - 1/p)$, the variance decomposes as

$$s_k^2 = V_1(k, M) + V_2(k, M)$$

with V_1 the diagonal variance contribution and V_2 the off-diagonal covariance sum. The diagonal V_1 is handled by partitioning primes by level $m_p = j$ and applying a partial-summation analogue of Mertens' second theorem that yields $\sum_{p: m_p=j} 1/p^2 \sim 1/(M \ln M)$ for each fixed j , giving $M \ln M \cdot V_1 \rightarrow B(c)$ by dominated convergence. The off-diagonal V_2 is bounded using the exact covariance identity

$$\beta_{\{p,q\}}(M) - q_p q_q = -\frac{m_p m_q}{(M-1)^2} \quad \text{for } pq > M,$$

together with a mean-value bound. Section 3 carries this out in detail.

Theorem 2 is proved via the Hall–Heyde martingale central limit theorem [5, Thm. 3.2], applied to the Doob filtration $\mathcal{F}_i = \sigma(X_1, \dots, X_i)$. Section 4 derives the martingale decomposition $W_k = \sum_{i=1}^k D_i$ and identifies the explicit form of the increments.

The application of [5] requires verifying two conditions. Section 5 proves the conditional Lindeberg condition via a sup-norm bound on the increments: $\max_i |D_i| = O((\log M)^3/M) = o(s_k)$ on a super-polynomially likely event. The bound exploits two facts simultaneously. First, a coupon-collector tail estimate: with super-polynomially high probability, every prime p with $m_p \geq (\log M)^2$ has been sampled by time $k/2$. Second, a sieve-geometric observation: once contributing primes are restricted to $p > M/(\log M)^2$, at most one of them can divide any given $x \leq M$, because $(M/(\log M)^2)^2 > M$ for M sufficiently large. Either fact alone gives only a weak bound; together they yield the required estimate.

Section 6 proves the conditional-variance concentration $\sum_{i=1}^k \mathbb{E}[D_i^2 \mid \mathcal{F}_{i-1}]/s_k^2 \xrightarrow{P} 1$. The argument rests on two explicit moment bounds. Lemma 14 controls the per-prime variance $\text{Var}(R_{pp}^{(k)})$ via a substitution $Y_p = q_p^{-2(T_p \wedge k)}$ that reduces the quantity to a function of a single geometric random variable. Lemma 15 bounds the cross-prime covariance $|\text{Cov}(R_{pp}^{(k)}, R_{qq}^{(k)})|$ for $pq > M$ via a comparison between the true joint law of the hitting times (T_p, T_q) and its independent product. These bounds are combined with a single-pair second-moment estimate, a level-summation argument, and Minkowski's inequality in L^2 , which controls all cross-covariances among the prime-pair summands without assuming orthogonality.

Section 7 assembles the pieces for Theorem 2. Section 8 presents numerical diagnostics for both theorems.

2 Preliminaries

Throughout, $M \geq 2$ is a fixed integer, $X_1, X_2, \dots$ are i.i.d. uniform on $\{2, \dots, M\}$, $L_k = \text{lcm}(X_1, \dots, X_k)$, and $\rho_k(M) = \mathbb{E}[\varphi(L_k)/L_k]$. For a prime $p \leq M$, set

$$m_p := \lfloor M/p \rfloor, \quad \pi_p := \frac{m_p}{M-1}, \quad q_p := 1 - \pi_p, \quad \xi_p^{(i)} := \mathbf{1}\{p \mid X_i\} - \pi_p.$$

Then m_p is the number of multiples of p in $\{2, \dots, M\}$ and $\pi_p = \mathbb{P}(p \mid X_i)$ for any single draw. For $T \subseteq \mathcal{P}_M := \{p : p \leq M\}$, let $A_T(M) = \{n \in \{2, \dots, M\} : \gcd(n, \prod_{p \in T} p) = 1\}$, so $\beta_T(M) = |A_T(M)|/(M-1)$ is the probability that a single uniform draw is coprime to every prime in T ; in

particular $\beta_\emptyset = 1$ and $\beta_{\{p\}} = q_p$. Let $T_p := \min\{i \geq 1 : p \mid X_i\}$, which is geometric with success probability π_p . Let $\mathcal{F}_i := \sigma(X_1, \dots, X_i)$ and $\mathcal{F}_0 := \{\emptyset, \Omega\}$. Logarithms are natural throughout, and $c > 0$ is fixed with $k := \lfloor cM \rfloor$ unless otherwise stated. We write C for an absolute constant and $C(c)$ for a constant depending only on c ; these may vary from line to line.

We record three elementary facts and the basic subset-density lemma.

Lemma 3 (Elementary subset-density bounds; [12, Lems. 2.1–2.4]). *For every $T \subseteq \mathcal{P}_M$:*

- (i) $\mathbb{P}(\text{no prime in } T \text{ divides } L_k) = \beta_T(M)^k$;
- (ii) $\beta_T(M) \leq q_p(M)$ for every $p \in T$, $T \neq \emptyset$;
- (iii) if $|T| = r \geq 1$, then $\beta_T(M)^k \leq \prod_{p \in T} q_p(M)^{k/r}$ for every real $k > 0$.

Lemma 4 (Rosser–Schoenfeld level envelope; [9], [12, Lem. 6.3]). *For all $M \geq 289$ and all $1 \leq j \leq M/17$,*

$$|\{p \leq M : m_p = j\}| \leq \frac{1.26(M/j)}{\ln(M/j)}, \quad \sum_{p: m_p=j} \frac{1}{p-1} \leq \frac{5.04}{\ln(M/j)}.$$

In particular, for $1 \leq j \leq M^{1/2}$,

$$|\{p \leq M : m_p = j\}| \leq \frac{2.52 M}{j \ln M}, \quad \sum_{p: m_p=j} \frac{1}{p-1} \leq \frac{10.08}{\ln M}.$$

Proposition 5 (Disjointness for $pq > M$; [12, Rem. 5.6]). *For distinct primes $p, q \leq M$ with $pq > M$, the events $\{p \mid X\}$ and $\{q \mid X\}$ are disjoint under a single uniform draw X on $\{2, \dots, M\}$; equivalently $\mathbb{P}(p \mid X, q \mid X) = 0$, so $\mathbb{P}(p \nmid X, q \nmid X) = 1 - \pi_p - \pi_q$ and*

$$\beta_{\{p,q\}}(M) - q_p(M) q_q(M) = -\pi_p \pi_q.$$

Part (iii) of Lemma 3 is the geometric-mean bound of [12, Lem. 2.4], which underlies the sharp joint-scaling result in the companion paper.

3 Variance asymptotic: proof of Theorem 1

Let $k = \lfloor cM \rfloor$. Write $\varphi(L_k)/L_k = \prod_{p \leq M} (1 - 1/p)^{X_p}$, where $X_p = \mathbf{1}\{p \mid L_k\}$, so

$$\ln(\varphi(L_k)/L_k) = \sum_{p \leq M} X_p \ln(1 - 1/p).$$

The variance of X_p is $(1 - q_p^k)q_p^k$ and for $p \neq q$,

$$\text{Cov}(X_p, X_q) = \mathbb{P}(p \mid L_k, q \mid L_k) - (1 - q_p^k)(1 - q_q^k) = \beta_{\{p,q\}}^k - q_p^k q_q^k,$$

as follows from the inclusion-exclusion identity $\mathbb{P}(p \mid L_k, q \mid L_k) = 1 - q_p^k - q_q^k + \beta_{\{p,q\}}^k$. Therefore

$$s_k^2 = V_1(k, M) + V_2(k, M), \tag{2}$$

where the diagonal and off-diagonal contributions are

$$\begin{aligned} V_1(k, M) &= \sum_{p \leq M} \ln(1 - 1/p)^2 q_p^k (1 - q_p^k), \\ V_2(k, M) &= \sum_{p \neq q, p, q \leq M} \ln(1 - 1/p) \ln(1 - 1/q) (\beta_{\{p,q\}}^k - q_p^k q_q^k). \end{aligned}$$

3.1 Diagonal contribution

We show $M \ln M \cdot V_1(k, M) \rightarrow B(c)$. Partition by $m_p = j$:

$$V_1(k, M) = \sum_{j \geq 1} (1 - j/(M-1))^k (1 - (1 - j/(M-1))^k) \sum_{p: m_p=j} \ln(1 - 1/p)^2.$$

For $p \in (M/(j+1), M/j]$, $\ln(1 - 1/p)^2 = 1/p^2 + O(1/p^3)$, and since $p \geq M/(j+1)$ we have $1/p^3 = O((j+1)^3/M^3)$. Hence

$$\sum_{p: m_p=j} \ln(1 - 1/p)^2 = \sum_{p: m_p=j} \frac{1}{p^2} + O\left(\frac{(j+1)^3}{M^3} \cdot \pi(M/j)\right) = \sum_{p: m_p=j} \frac{1}{p^2} + O\left(\frac{(j+1)^2}{M^2 \ln M}\right).$$

Lemma 6 (Level- j sum of reciprocal squares). *For each fixed $j \geq 1$,*

$$M \ln M \cdot \sum_{p: m_p=j} \frac{1}{p^2} \longrightarrow 1 \quad (M \rightarrow \infty).$$

Proof. By partial summation and the prime number theorem in its standard quantitative form [11], $\sum_{p \leq x} 1/p^2 = \pi(x)/x^2 + 2 \int_2^x \pi(t)/t^3 dt$, so $\sum_{p \in (A, B]} 1/p^2 = \pi(B)/B^2 - \pi(A)/A^2 + 2 \int_A^B \pi(t)/t^3 dt$. With $A = M/(j+1)$, $B = M/j$, and using $\pi(t) = t/\ln t + O(t/(\ln t)^2)$,

$$\pi(B)/B^2 = \frac{1}{B \ln B} (1 + O(1/\ln B)) = \frac{j}{M \ln(M/j)} (1 + o(1)) = \frac{j}{M \ln M} (1 + o(1)),$$

and analogously $\pi(A)/A^2 = (j+1)/(M \ln M) (1 + o(1))$. Their difference is

$$\pi(B)/B^2 - \pi(A)/A^2 = -\frac{1}{M \ln M} (1 + o(1)).$$

The integral is

$$2 \int_A^B \frac{1}{t^2 \ln t} (1 + O(1/\ln t)) dt = \frac{2}{\ln M} \int_A^B \frac{dt}{t^2} (1 + o(1)) = \frac{2}{\ln M} \left(\frac{1}{A} - \frac{1}{B} \right) (1 + o(1)),$$

and $1/A - 1/B = (j+1)/M - j/M = 1/M$, so the integral contributes $2/(M \ln M) (1 + o(1))$. Summing the two contributions:

$$\sum_{p: m_p=j} \frac{1}{p^2} = \left[-\frac{1}{M \ln M} + \frac{2}{M \ln M} \right] (1 + o(1)) = \frac{1}{M \ln M} (1 + o(1)),$$

for each fixed $j \geq 1$. □

Dominated convergence. For each fixed j , Lemma 6 together with $(1 - j/(M-1))^k \rightarrow e^{-cj}$ and $(1 - (1 - j/(M-1))^k) \rightarrow 1 - e^{-cj}$ gives

$$M \ln M \cdot (1 - j/(M-1))^k (1 - (1 - j/(M-1))^k) \sum_{p: m_p=j} \ln(1 - 1/p)^2 \longrightarrow e^{-cj} (1 - e^{-cj}).$$

A uniform envelope follows directly from the prime-count part of Lemma 4. If $m_p = j$, then $p > M/(j+1)$ and hence $p^{-2} < (j+1)^2/M^2$. Therefore, for $1 \leq j \leq M/17$ and $M \geq 289$,

$$\sum_{p: m_p=j} \frac{1}{p^2} \leq |\{p: m_p=j\}| \frac{(j+1)^2}{M^2} \leq \frac{1.26(M/j)}{\ln(M/j)} \frac{(j+1)^2}{M^2} = \frac{1.26(j+1)^2}{Mj \ln(M/j)}.$$

For $j \leq M^{1/2}$, $\ln(M/j) \geq \frac{1}{2} \ln M$, hence $M \ln M \cdot \sum_{p: m_p=j} 1/p^2 \leq 2.52 (j+1)^2/j$ for $j \geq 1$ (a function of j of order j). Multiplied by the bound $(1 - j/(M-1))^k \leq 2e^{-cj}$ for the other factors, each term is bounded by $2.52 (j+1)^2/j \cdot 2e^{-cj} = 5.04 (j+1)^2/j \cdot e^{-cj}$, which is summable. The tail $j > M^{1/2}$ decays super-exponentially since $(1 - j/(M-1))^k \leq e^{-c\sqrt{M}/2}$ there and the prime count is $O(M)$. By dominated convergence,

$$\lim_{M \rightarrow \infty} M \ln M \cdot V_1(k, M) = \sum_{j \geq 1} e^{-cj} (1 - e^{-cj}) = B(c). \quad (3)$$

3.2 Off-diagonal contribution is negligible

We show $M \ln M \cdot V_2(k, M) \rightarrow 0$. The starting point is an exact identity valid in a restricted range of prime pairs.

Lemma 7 (Covariance identity for disjoint prime levels). *Let $p, q \in \mathcal{P}_M$ with $p \neq q$ and $pq > M$. Then*

$$\beta_{\{p,q\}}(M) - q_p(M) q_q(M) = -\frac{m_p m_q}{(M-1)^2}. \quad (4)$$

Proof. By inclusion-exclusion, $|A_{\{p,q\}}(M)| = M - \lfloor M/p \rfloor - \lfloor M/q \rfloor + \lfloor M/(pq) \rfloor - 1$. When $pq > M$, $\lfloor M/(pq) \rfloor = 0$, so $(M-1)\beta_{\{p,q\}} = M-1 - m_p - m_q$. Hence

$$(M-1)^2(\beta_{\{p,q\}} - q_p q_q) = (M-1)(M-1 - m_p - m_q) - (M-1 - m_p)(M-1 - m_q) = -m_p m_q. \quad \square$$

Remark 8. For levels (j_1, j_2) with $(j_1+1)(j_2+1) < M$, every prime p with $m_p = j_1$ and q with $m_q = j_2$ satisfies $pq \geq M/(j_1+1) \cdot M/(j_2+1) = M^2/((j_1+1)(j_2+1)) > M$, so (4) applies. In particular, if $\max(j_1, j_2) \leq M^{1/4}$ then $(j_1+1)(j_2+1) \leq (M^{1/4}+1)^2 < M$ for all $M \geq 7$, so the identity applies to every pair in this range.

Main-range bound. Let $\mathcal{J}(M) = \{(p, q) : p \neq q, p, q \leq M, \max(m_p, m_q) \leq M^{1/4}\}$ and write

$$V_2^{\text{main}}(k, M) := \sum_{(p,q) \in \mathcal{J}(M)} \ln(1 - 1/p) \ln(1 - 1/q) (\beta_{\{p,q\}}^k - q_p^k q_q^k),$$

with $V_2^{\text{tail}}(k, M) := V_2(k, M) - V_2^{\text{main}}(k, M)$.

For $(p, q) \in \mathcal{J}(M)$, let $j_1 = m_p$, $j_2 = m_q$. By Remark 8 and Lemma 7, $\beta_{\{p,q\}} - q_p q_q = -j_1 j_2 / (M-1)^2$; in particular $\beta_{\{p,q\}} < q_p q_q$, so by the mean-value form $|a^k - b^k| \leq k|a - b| \max(a, b)^{k-1}$ with $a = \beta_{\{p,q\}}$, $b = q_p q_q$,

$$|\beta_{\{p,q\}}^k - q_p^k q_q^k| \leq \frac{k j_1 j_2}{(M-1)^2} (q_p q_q)^{k-1}. \quad (5)$$

Since $q_p q_q \geq 1 - (j_1 + j_2)/(M-1) \geq 1/2$ for M large enough (because $j_1 + j_2 \leq 2M^{1/4} = o(M)$), we have $(q_p q_q)^{k-1} \leq 2(q_p q_q)^k$. Using also $|\ln(1 - 1/p)| \leq 1/(p-1)$, the contribution of any pair $(p, q) \in \mathcal{J}(M)$ with $m_p = j_1$, $m_q = j_2$ is at most

$$\frac{1}{(p-1)(q-1)} \cdot \frac{2k j_1 j_2}{(M-1)^2} (q_p q_q)^k.$$

Summing over pairs at fixed levels (j_1, j_2) — noting $(q_p q_q)^k$ depends on the pair only through j_1, j_2 — and using $S_j(M) := \sum_{p: m_p=j} 1/(p-1)$,

$$\sum_{\substack{(p,q) \in \mathcal{J}(M) \\ m_p=j_1, m_q=j_2}} \frac{1}{(p-1)(q-1)} \leq S_{j_1}(M) S_{j_2}(M).$$

By Lemma 4, $S_j(M) \leq 10.08 / \ln M$ uniformly for $1 \leq j \leq M^{1/2}$ and $M \geq 289$, hence certainly for $j \leq M^{1/4}$. Therefore

$$|V_2^{\text{main}}(k, M)| \leq \frac{2k}{(M-1)^2} \cdot \frac{(10.08)^2}{(\ln M)^2} \cdot \sum_{j_1, j_2 \geq 1} j_1 j_2 e^{-c(j_1+j_2)} (1+o(1)),$$

where the factor $(1+o(1))$ reflects $(q_p q_q)^k = e^{-c(j_1+j_2)}(1+o(1))$ in the joint regime (uniformly over $j_1, j_2 \leq M^{1/4}$, by the same argument as in Section 3.1). The double sum factorises:

$$\sum_{j_1, j_2 \geq 1} j_1 j_2 e^{-c(j_1+j_2)} = \left(\sum_{j \geq 1} j e^{-cj} \right)^2 = \frac{e^{2c}}{(e^c - 1)^4}.$$

With $k = \lfloor cM \rfloor$ so $k/(M-1)^2 = O(1/M)$,

$$|V_2^{\text{main}}(k, M)| = O\left(\frac{1}{M(\ln M)^2}\right). \quad (6)$$

Tail-range bound. For $(p, q) \notin \mathcal{J}(M)$, at least one of m_p, m_q exceeds $M^{1/4}$. Say $m_p > M^{1/4}$; then $q_p^k \leq (1 - M^{1/4}/(M-1))^{\lfloor cM \rfloor} \leq \exp(-cM^{1/4}(1+o(1)))$. Using Lemma 3(ii) in the form $\beta_{\{p,q\}} \leq q_p$, $\beta_{\{p,q\}}^k \leq q_p^k$, and the trivial bound $q_p^k q_q^k \leq q_p^k$,

$$|\beta_{\{p,q\}}^k - q_p^k q_q^k| \leq 2q_p^k.$$

Writing $\Lambda(M) := \sum_{p \leq M} 1/(p-1) = O(\ln \ln M)$ (Mertens),

$$\begin{aligned} |V_2^{\text{tail}}(k, M)| &\leq 2 \sum_{p: m_p > M^{1/4}} \frac{q_p^k}{p-1} \sum_{q \leq M, q \neq p} \frac{1}{q-1} + (\text{symmetric term with } p, q \text{ swapped}) \\ &\leq 4 \Lambda(M) \exp(-cM^{1/4}(1+o(1))) \Lambda(M) \\ &= O((\ln \ln M)^2 e^{-cM^{1/4}}), \end{aligned}$$

which vanishes super-polynomially in M . In particular $M \ln M \cdot V_2^{\text{tail}} \rightarrow 0$.

Combining (6) and the tail bound,

$$|V_2(k, M)| = O\left(\frac{1}{M(\ln M)^2}\right), \quad (7)$$

so $M \ln M \cdot |V_2(k, M)| = O(1/\ln M) \rightarrow 0$. Combined with (3), this gives (1). $\square$

Remark 9 (Sign of V_2). The leading contribution to V_2 comes from the region $\max(m_p, m_q) \leq M^{1/4}$. There $\beta_{\{p,q\}} - q_p q_q = -m_p m_q / (M-1)^2 < 0$, so $\beta_{\{p,q\}}^k - q_p^k q_q^k < 0$. The weights $\ln(1-1/p) \ln(1-1/q)$ are positive. Hence the main-region sum, and therefore V_2 for M large, is strictly negative. Numerically (Section 8) V_2 is observed negative at every tested (k, M) .

Remark 10 ($s_k \asymp (M \ln M)^{-1/2}$). Theorem 1 gives $s_k^2 \sim B(c)/(M \ln M)$ with $B(c) > 0$ for every $c > 0$, so in particular $s_k \asymp (M \ln M)^{-1/2}$ in the joint regime. This scale is used throughout the CLT proof below.

4 Martingale decomposition

Let $F_k := \ln(\varphi(L_k)/L_k) = \sum_{p \leq M} \ln(1 - 1/p) \mathbf{1}\{p \mid L_k\}$ and $D_i := \mathbb{E}[F_k \mid \mathcal{F}_i] - \mathbb{E}[F_k \mid \mathcal{F}_{i-1}]$ for $1 \leq i \leq k$.

Theorem 11 (Doob decomposition). *$(D_i, \mathcal{F}_i)_{i=1}^k$ is a martingale-difference sequence with $\sum_{i=1}^k D_i = W_k$, and*

$$D_i = \sum_{\substack{p \leq M \\ p \nmid L_{i-1}}} \ln(1 - 1/p) q_p^{k-i} \xi_p^{(i)}. \quad (8)$$

Proof. By linearity, $\mathbb{E}[F_k \mid \mathcal{F}_i] = \sum_p \ln(1 - 1/p) \mathbb{E}[\mathbf{1}\{p \mid L_k\} \mid \mathcal{F}_i]$. Since $p \mid L_k$ iff $p \mid L_i$ or $p \mid X_j$ for some $j > i$, and $X_{i+1}, \dots, X_k \perp \mathcal{F}_i$,

$$\mathbb{E}[\mathbf{1}\{p \mid L_k\} \mid \mathcal{F}_i] = \mathbf{1}\{p \mid L_i\} + \mathbf{1}\{p \nmid L_i\}(1 - q_p^{k-i}).$$

On $\{p \mid L_{i-1}\}$, both $\mathbb{E}[\mathbf{1}\{p \mid L_k\} \mid \mathcal{F}_i]$ and $\mathbb{E}[\mathbf{1}\{p \mid L_k\} \mid \mathcal{F}_{i-1}]$ equal 1, so this prime contributes nothing to D_i . On $\{p \nmid L_{i-1}\}$, case analysis on $\mathbf{1}\{p \mid X_i\} \in \{0, 1\}$ gives

$$\mathbb{E}[\mathbf{1}\{p \mid L_k\} \mid \mathcal{F}_i] - \mathbb{E}[\mathbf{1}\{p \mid L_k\} \mid \mathcal{F}_{i-1}] = \begin{cases} q_p^{k-i+1}, & p \mid X_i, \\ -q_p^{k-i}(1 - q_p), & p \nmid X_i, \end{cases}$$

which equals $q_p^{k-i} \xi_p^{(i)}$ in either case. Summing over p gives (8); martingale-difference and telescoping properties are immediate. $\square$

By orthogonality of martingale increments and Theorem 1, $\sum_{i=1}^k \mathbb{E}[D_i^2] = s_k^2 \sim B(c)/(M \ln M)$.

5 Conditional Lindeberg condition

Let $i_0 := \lceil k/2 \rceil$ and define the event

$$\Omega^* := \bigcap_{\substack{p \leq M \\ m_p \geq (\log M)^2}} \{p \mid L_{i_0-1}\}.$$

Theorem 12 (Uniform increment bound). *For every fixed $c > 0$ and every $K \geq 0$, $\mathbb{P}((\Omega^*)^c) = o(M^{-K})$ as $M \rightarrow \infty$. Moreover, there exists $M_0 = M_0(c)$ such that, for $M \geq M_0$,*

$$\max_{1 \leq i \leq k} |D_i| \leq C(c) (\log M)^3 / M = o(s_k) \quad (9)$$

on Ω^* . More precisely, for $i < i_0$ the bound holds almost surely, while for $i \geq i_0$ it holds uniformly over the possible value of X_i whenever the history belongs to Ω^* .

Proof. Probability of Ω^* . For a prime p with $m_p \geq (\log M)^2$,

$$\mathbb{P}(p \nmid L_{i_0-1}) = q_p^{i_0-1} \leq \exp(-\pi_p(i_0 - 1)) \leq \exp(-(c/2)(\log M)^2(1 + o(1))).$$

A union bound over at most $\pi(M) \leq M/\ln M$ primes gives

$$\mathbb{P}((\Omega^*)^c) \leq \frac{M}{\ln M} \exp(-(c/2)(\log M)^2(1 + o(1))) = o(M^{-K})$$

for every K .

Case A: $i \geq i_0$, on Ω^* . Decompose $D_i = S_i(X_i) - C_i$, where

$$S_i(x) := \sum_{\substack{p|x \\ p \nmid L_{i-1}}} \ln(1 - 1/p) q_p^{k-i}, \quad C_i := \sum_{p \nmid L_{i-1}} \ln(1 - 1/p) q_p^{k-i} \pi_p.$$

On Ω^* , every contributing prime satisfies $m_p < (\log M)^2$, hence $p > M/(\log M)^2$. For $M \geq M_0(c)$ large enough that $M > (\log M)^4$, any two such primes p_1, p_2 obey

$$p_1 p_2 > (M/(\log M)^2)^2 = M^2/(\log M)^4 > M,$$

so p_1 and p_2 cannot both divide any $x \in \{2, \dots, M\}$. Hence at most one contributing prime divides x , giving

$$|S_i(x)| \leq \max_{\substack{p \nmid L_{i-1} \\ m_p < (\log M)^2}} \frac{q_p^{k-i}}{p-1} \leq \frac{(\log M)^2}{M-1}.$$

For C_i , a prime at level $j \leq (\log M)^2$ contributes at most $|\ln(1 - 1/p)| \pi_p \leq 1/(p-1) \cdot j/(M-1) \leq 2(j+1)/M \cdot (j+1)/M \leq 2(j+1)^2/M^2$ (using $q_p^{k-i} \leq 1$, the crudest factor since $k-i$ may be as small as zero when $i = k$). Summing using $|\{p : m_p = j\}| \leq 1.26 M/(j \ln(M/j))$ of Lemma 4:

$$|C_i| \leq \sum_{j=1}^{(\log M)^2} \frac{1.26 M}{j \ln(M/j)} \cdot \frac{2(j+1)^2}{M^2} \leq \frac{5.04}{M \ln M} \sum_{j=1}^{(\log M)^2} \frac{(j+1)^2}{j} = O\left(\frac{(\log M)^3}{M}\right),$$

since $\sum_{j=1}^J (j+1)^2/j = \sum_j (j+2+1/j) = O(J^2)$ at $J = (\log M)^2$.

Combining, $|D_i| \leq |S_i(X_i)| + |C_i| \leq C(c)(\log M)^3/M$ uniformly over X_i on Ω^* .

Case B: $i < i_0$, unconditionally. Put $r := k - i$. Since $i < \lceil k/2 \rceil$, one has $r > k/2$; hence, for all sufficiently large M , $r \geq cM/3$. We again write $D_i = S_i(X_i) - C_i$.

For the part of $S_i(x)$ coming from primes $p \geq \sqrt{M}$, at most one such prime can divide a fixed $x \leq M$: two distinct prime divisors in this range would have product $> M$. If that prime has level $m_p = j \leq \sqrt{M}$, then

$$\frac{q_p^r}{p-1} \leq \frac{2(j+1)}{M} \exp\left(-\frac{rj}{M-1}\right) \leq \frac{2(j+1)}{M} e^{-cj/4} = O_c(M^{-1}),$$

uniformly in j . For primes $p < \sqrt{M}$, one has $m_p \geq \sqrt{M} - 1$, so

$$q_p^r \leq \exp\left(-\frac{rm_p}{M-1}\right) \leq e^{-c\sqrt{M}/4}$$

for large M . Consequently their total contribution is at most $e^{-c\sqrt{M}/4} \sum_{p \leq M} (p-1)^{-1} = O(e^{-c\sqrt{M}/4} \log \log M)$. Thus $\sup_x |S_i(x)| = O_c(M^{-1})$.

For the centring term, split by levels. When $j \leq \sqrt{M}$, Lemma 4, $q_p^r \leq e^{-cj/4}$, $\pi_p = j/(M-1)$, and $(p-1)^{-1} \leq 2(j+1)/M$ give

$$\sum_{p: m_p = j} |\ln(1 - 1/p)| \pi_p q_p^r \leq \frac{C}{M \ln M} (j+1) e^{-cj/4}.$$

Summing over j yields $O_c((M \ln M)^{-1})$. The levels $j > \sqrt{M}$ contribute $O(e^{-c\sqrt{M}/4} \log \log M)$. Therefore $|C_i| = O_c((M \ln M)^{-1})$ and $|D_i| = O_c(M^{-1})$ uniformly over all histories and all values of X_i .

Comparison to s_k . $(\log M)^3/M \cdot (M \ln M)^{1/2} = (\log M)^{7/2}/\sqrt{M} \rightarrow 0$, establishing (9). $\square$

Corollary 13 (Conditional Lindeberg). *For every $\varepsilon > 0$,*

$$\frac{1}{s_k^2} \sum_{i=1}^k \mathbb{E}[D_i^2 \mathbf{1}\{|D_i| > \varepsilon s_k\} \mid \mathcal{F}_{i-1}] \xrightarrow{P} 0.$$

Proof. By Remark 10, $s_k \asymp (M \ln M)^{-1/2}$, and hence $(\log M)^3/M = o(s_k)$. For M sufficiently large, Theorem 12 implies that every summand in the conditional Lindeberg expression vanishes on Ω^* : for $i < i_0$ this holds unconditionally, and for $i \geq i_0$ it holds uniformly over X_i conditional on any history in Ω^* . Therefore the entire nonnegative conditional Lindeberg sum is zero on Ω^* . Consequently, for every $\delta > 0$,

$$\mathbb{P}\left(\frac{1}{s_k^2} \sum_{i=1}^k \mathbb{E}[D_i^2 \mathbf{1}\{|D_i| > \varepsilon s_k\} \mid \mathcal{F}_{i-1}] > \delta\right) \leq \mathbb{P}((\Omega^*)^c) \rightarrow 0.$$

$\square$

6 Conditional-variance concentration

Let $U_i := \mathbb{E}[D_i^2 \mid \mathcal{F}_{i-1}]$. Expanding via (8),

$$\sum_{i=1}^k U_i = A + 2B$$

where

$$A := \sum_{p \leq M} c_{pp} R_{pp}^{(k)}, \quad B := \sum_{p < q \leq M} c_{pq} R_{pq}^{(k)},$$

with $c_{pq} := \ln(1 - 1/p) \ln(1 - 1/q) \kappa_{pq}$, $\kappa_{pp} := \pi_p q_p$, $\kappa_{pq} := \beta_{\{p,q\}} - q_p q_q$ for $p \neq q$, and

$$R_{pp}^{(k)} := \sum_{i=1}^k \mathbf{1}\{T_p \geq i\} q_p^{2(k-i)}, \quad R_{pq}^{(k)} := \sum_{i=1}^k \mathbf{1}\{T_p \geq i, T_q \geq i\} (q_p q_q)^{k-i}.$$

By Proposition 5, $\kappa_{pq} = -\pi_p \pi_q$ whenever $pq > M$.

The main estimates of this section are the following.

Lemma 14 (Per-prime variance). *For every prime $p \leq M$ with $q_p \geq 1/2$ and every $k \geq 1$,*

$$\text{Var}(R_{pp}^{(k)}) \leq \frac{3 q_p^k}{\pi_p^2}. \quad (10)$$

Lemma 15 (Cross-prime covariance). *There exists a constant $C_0(c) > 0$ (depending only on c) such that, for distinct primes $p, q \leq M$ with $pq > M$ and $\pi_p + \pi_q \leq 1/3$, in the joint regime $k = \lfloor cM \rfloor$,*

$$|\text{Cov}(R_{pp}^{(k)}, R_{qq}^{(k)})| \leq C_0(c) k q_p^k q_q^k. \quad (11)$$

An admissible value is $C_0(c) = 57 + 54/c$; this is what the proof chain yields directly, though numerical evidence suggests that it is not sharp. Only the shape of the bound — linear in k , multiplicative in $q_p^k q_q^k$ — is used in Theorem 17 below.

6.1 Proof of Lemma 14

Set $Y_p := q_p^{-2(T_p \wedge k)}$. Then $R_{pp}^{(k)} = q_p^{2k}(Y_p - 1)/(1 - q_p^2)$, and

$$\text{Var}(R_{pp}^{(k)}) = \frac{q_p^{4k} \text{Var}(Y_p)}{(1 - q_p^2)^2} \leq \frac{q_p^{4k} \mathbb{E}[Y_p^2]}{(1 - q_p^2)^2}.$$

Compute

$$\mathbb{E}[Y_p^2] = \sum_{m=1}^k q_p^{-4m} \pi_p q_p^{m-1} + q_p^{-3k} = \frac{\pi_p}{q_p} \cdot \frac{q_p^{-3k} - 1}{1 - q_p^3} + q_p^{-3k}.$$

Since $1 - q_p^3 = \pi_p(1 + q_p + q_p^2)$, the first term simplifies to $(q_p^{-3k} - 1)/(q_p(1 + q_p + q_p^2)) \leq q_p^{-3k}/(q_p(1 + q_p + q_p^2))$. For $q_p \geq 1/2$, $q_p(1 + q_p + q_p^2) \geq (1/2)(1 + 1/2 + 1/4) = 7/8$, so

$$\mathbb{E}[Y_p^2] \leq q_p^{-3k}(8/7 + 1) = (15/7) q_p^{-3k} \leq 3 q_p^{-3k}.$$

Using $(1 - q_p^2)^2 = \pi_p^2(1 + q_p)^2 \geq \pi_p^2$:

$$\text{Var}(R_{pp}^{(k)}) \leq \frac{q_p^{4k} \cdot 3 q_p^{-3k}}{\pi_p^2} = \frac{3 q_p^k}{\pi_p^2}.$$

6.2 Proof of Lemma 15

Write $\mathbb{P}_{\text{dep}}$ for the joint law of (T_p, T_q) under the true distribution and $\mathbb{P}_{\text{ind}}$ for the product of the (common) marginals. By Proposition 5, at each step three outcomes are possible with probabilities β, π_p, π_q (where $\beta := \beta_{\{p,q\}} = 1 - \pi_p - \pi_q$), giving, for $1 \leq i < j \leq k$,

$$\mathbb{P}_{\text{dep}}(T_p = i, T_q = j) = \beta^{i-1} \pi_p q_q^{j-i-1} \pi_q,$$

and the analogous formula for $j < i$, with $\mathbb{P}_{\text{dep}}(T_p = T_q < \infty) = 0$ (disjointness). Under $\mathbb{P}_{\text{dep}}$, the marginal law of T_p is geometric with parameter π_p (at each step the event $\{p \mid X\}$ has probability π_p), and similarly for T_q . The product law with identical marginals is then

$$\mathbb{P}_{\text{ind}}(T_p = i, T_q = j) = \pi_p q_q^{i-1} \cdot \pi_q q_q^{j-1} = \pi_p \pi_q (q_p q_q)^{i-1} q_q^{j-i} \quad (i < j),$$

and analogously for $j < i$ and for the diagonal $\mathbb{P}_{\text{ind}}(T_p = T_q = m) = \pi_p \pi_q (q_p q_q)^{m-1}$. Set $\delta := \pi_p \pi_q / (q_p q_q)$, so $\beta / (q_p q_q) = 1 - \delta$. Under $\pi_p + \pi_q \leq 1/3$, $q_p q_q \geq (1 - \pi_p)(1 - \pi_q) \geq 1 - (\pi_p + \pi_q) \geq 2/3$, whence $\delta \leq (3/2)\pi_p \pi_q$ and $1/q_p, 1/q_q \leq 3/2$. Since marginals agree, $\text{Cov}_{\text{dep}}(Y_p, Y_q) = \mathbb{E}_{\text{dep}}[Y_p Y_q] - \mathbb{E}_{\text{ind}}[Y_p Y_q]$, where $Y_p := q_p^{-2(T_p \wedge k)}$, $Y_q := q_q^{-2(T_q \wedge k)}$.

Partition $\{(i, j) : 1 \leq i, j \leq k+1\}$ (with $T > k$ encoded as $T = k+1$) into the regions $i < j$, $i > j$, $i = j \leq k$, and $i = j = k+1$.

Region $i < j$. The density ratio is

$$\frac{\mathbb{P}_{\text{dep}}(T_p = i, T_q = j)}{\mathbb{P}_{\text{ind}}(T_p = i, T_q = j)} = \frac{(1 - \delta)^{i-1}}{q_q}.$$

Write $(1 - \delta)^{i-1}/q_q - 1 = [(1 - \delta)^{i-1} - 1]/q_q + \pi_q/q_q$. Using $|(1 - \delta)^{i-1} - 1| \leq (i-1)\delta$,

$$\begin{aligned} & |\mathbb{E}_{\text{dep}}[Y_p Y_q; i < j] - \mathbb{E}_{\text{ind}}[Y_p Y_q; i < j]| \\ & \leq (\delta/q_q) \mathbb{E}_{\text{ind}}[(T_p - 1)Y_p Y_q; i < j] + (\pi_q/q_q) \mathbb{E}_{\text{ind}}[Y_p Y_q; i < j]. \end{aligned}$$

Under $\mathbb{P}_{\text{ind}}$, independence gives $\mathbb{E}[T_p Y_p Y_q; i < j] \leq \mathbb{E}[T_p Y_p] \mathbb{E}[Y_q]$. For $r = q_p^{-1} > 1$: $\sum_{m=1}^k m r^m \leq k r^{k+1} / (r - 1)$ (telescoping identity $(r - 1) \sum_{m=1}^k m r^m = k r^{k+1} - \sum_{m=1}^k r^m$, dropping the positive term). Hence

$$\mathbb{E}[T_p Y_p] = \frac{\pi_p}{q_p} \sum_{m=1}^k m q_p^{-m} + (k+1) q_p^{-k} \leq \frac{\pi_p}{q_p} \cdot \frac{k q_p^{-k-1}}{\pi_p} + 2k q_p^{-k} = k q_p^{-k-1} + 2k q_p^{-k} \leq 4k q_p^{-k},$$

for $q_p \geq 1/2$. Analogously $\mathbb{E}[Y_q] \leq 3q_q^{-k}$ (from $\mathbb{E}[Y_q] = (1 + 1/q_q)q_q^{-k} - 1/q_q \leq 3q_q^{-k}$) and $\mathbb{E}[Y_p] \leq 3q_p^{-k}$. Thus

$$\mathbb{E}_{\text{ind}}[T_p Y_p Y_q; i < j] \leq 12k q_p^{-k} q_q^{-k}, \quad \mathbb{E}_{\text{ind}}[Y_p Y_q; i < j] \leq 9 q_p^{-k} q_q^{-k}.$$

Using $\delta/q_q \leq (3/2)(3/2)\pi_p \pi_q = (9/4)\pi_p \pi_q$ and $\pi_q/q_q \leq (3/2)\pi_q$:

$$|\mathbb{E}_{\text{dep}}[Y_p Y_q; i < j] - \mathbb{E}_{\text{ind}}[Y_p Y_q; i < j]| \leq 27 k \pi_p \pi_q q_p^{-k} q_q^{-k} + \frac{27}{2} \pi_q q_p^{-k} q_q^{-k}. \quad (12)$$

Region $i > j$. By symmetry (swap roles of p and q):

$$|\mathbb{E}_{\text{dep}}[Y_p Y_q; i > j] - \mathbb{E}_{\text{ind}}[Y_p Y_q; i > j]| \leq 27 k \pi_p \pi_q q_p^{-k} q_q^{-k} + \frac{27}{2} \pi_p q_p^{-k} q_q^{-k}. \quad (13)$$

Region $i = j \leq k$. Here $\mathbb{P}_{\text{dep}} = 0$ and $\mathbb{P}_{\text{ind}}(T_p = T_q = m) = \pi_p \pi_q (q_p q_q)^{m-1}$, contributing

$$\mathbb{E}_{\text{ind}}[Y_p Y_q; T_p = T_q \leq k] = \pi_p \pi_q \sum_{m=1}^k (q_p q_q)^{m-1} (q_p q_q)^{-2m} = \pi_p \pi_q (q_p q_q)^{-1} \sum_{m=1}^k (q_p q_q)^{-m},$$

and using $\sum_{m=1}^k r^m \leq k r^k$ for $r = (q_p q_q)^{-1} \geq 1$,

$$\mathbb{E}_{\text{ind}}[Y_p Y_q; T_p = T_q \leq k] \leq \pi_p \pi_q k (q_p q_q)^{-k-1} \leq \frac{3}{2} k \pi_p \pi_q q_p^{-k} q_q^{-k}. \quad (14)$$

Region $i = j = k + 1$. Here $\mathbb{P}_{\text{dep}}(T_p > k, T_q > k) = \beta^k$ and $\mathbb{P}_{\text{ind}}(T_p > k, T_q > k) = (q_p q_q)^k$, with $Y_p Y_q = q_p^{-2k} q_q^{-2k}$:

$$|\mathbb{E}_{\text{dep}}[Y_p Y_q; \text{both} > k] - \mathbb{E}_{\text{ind}}[Y_p Y_q; \text{both} > k]| = (q_p q_q)^{-2k} |\beta^k - (q_p q_q)^k|.$$

Using $|\beta^k - (q_p q_q)^k| = (q_p q_q)^k |1 - (1 - \delta)^k| \leq (q_p q_q)^k k \delta$ and $\delta \leq (3/2)\pi_p \pi_q$,

$$|\mathbb{E}_{\text{dep}}[Y_p Y_q; \text{both} > k] - \mathbb{E}_{\text{ind}}[Y_p Y_q; \text{both} > k]| \leq \frac{3}{2} k \pi_p \pi_q (q_p q_q)^{-k} \leq \frac{3}{2} k \pi_p \pi_q q_p^{-k} q_q^{-k}. \quad (15)$$

Assembly. Combining (12)–(15),

$$|\text{Cov}(Y_p, Y_q)| \leq 57 k \pi_p \pi_q q_p^{-k} q_q^{-k} + \frac{27}{2} (\pi_p + \pi_q) q_p^{-k} q_q^{-k}. \quad (16)$$

(The coefficient $57 = 27 + 27 + 3/2 + 3/2$ dominates the individual contributions.) Using the linear relation $R_{pp}^{(k)} = q_p^{2k}(Y_p - 1)/(1 - q_p^2)$ from the proof of Lemma 14 (analogously for q) and $(1 - q_p^2)(1 - q_q^2) = \pi_p \pi_q (1 + q_p)(1 + q_q) \geq \pi_p \pi_q$:

$$|\text{Cov}(R_{pp}^{(k)}, R_{qq}^{(k)})| = \frac{(q_p q_q)^{2k} |\text{Cov}(Y_p, Y_q)|}{(1 - q_p^2)(1 - q_q^2)} \leq \frac{|\text{Cov}(Y_p, Y_q)| (q_p q_q)^{2k}}{\pi_p \pi_q}.$$

Substituting (16):

$$|\text{Cov}(R_{pp}^{(k)}, R_{qq}^{(k)})| \leq \left[57k + \frac{27}{2} \left(\frac{1}{\pi_q} + \frac{1}{\pi_p} \right) \right] q_p^k q_q^k.$$

Since $\pi_p \geq 1/(M - 1)$ trivially and $k = \lfloor cM \rfloor \geq cM - 1$, we have $M \leq (k + 1)/c$, so $1/\pi_p \leq M - 1 \leq (k + 1)/c$, and likewise for q . Therefore $1/\pi_p + 1/\pi_q \leq 2(k + 1)/c$, and using $k + 1 \leq 2k$ for $k \geq 1$,

$$|\text{Cov}(R_{pp}^{(k)}, R_{qq}^{(k)})| \leq [57 + 54/c] k q_p^k q_q^k.$$

Thus $C_0(c) := 57 + 54/c$ suffices for all $k \geq 1$. $\square$

6.3 Level-summation bound and assembly

Lemma 16 (Level-summation upper bound). *Let $f: \mathbb{N}_{\geq 1} \rightarrow \mathbb{R}_{\geq 0}$ satisfy $f(j) \leq A(1+j)^r$ for $A \geq 0$, $r \geq 0$. For each $c > 0$ there exist $M_0(c, r)$ and $C(c, r)$ such that for $M \geq M_0(c, r)$,*

$$L_M(f) := \sum_{p \leq M} \frac{f(m_p) q_p^{\lfloor cM \rfloor}}{(p-1)^4} \leq \frac{C(c, r) A}{M^3 \ln M}. \quad (17)$$

Proof. Partitioning by level and bounding $1/(p-1)^4 \leq 16(j+1)^4/M^4$ for $M \geq 2(j+1)$, together with Lemma 4:

$$S_j(M) := \sum_{p: m_p=j} \frac{1}{(p-1)^4} \leq \frac{20.16(j+1)^4}{j \ln(M/j) M^3}.$$

For $j \leq M^{1/2}$, $\ln(M/j) \geq (\ln M)/2$, so $S_j(M) \leq 40.32(j+1)^4/(j \ln M \cdot M^3)$.

For the exponential factor, $\ln(1-x) \leq -x$ (concavity) gives $\ln(1-j/(M-1))^{\lfloor cM \rfloor} \leq -cj + O(j/M)$ uniformly for $j \leq M^{1/2}$, so $(1-j/(M-1))^{\lfloor cM \rfloor} \leq 2e^{-cj}$ for $M \geq M_1(c)$.

Main range $j \leq M^{1/2}$:

$$\sum_{j=1}^{\lfloor M^{1/2} \rfloor} f(j) \cdot (1-j/(M-1))^{\lfloor cM \rfloor} S_j(M) \leq \frac{80.64}{M^3 \ln M} \sum_{j \geq 1} \frac{A(1+j)^{r+4} e^{-cj}}{j},$$

and the series converges to a finite $\Sigma(c, r) < \infty$.

Tail $j > M^{1/2}$: $q_p^k \leq e^{-c\sqrt{M}/2}$ times $\sum_p 1/(p-1)^4 = O(1)$ times $\sum_{j > M^{1/2}}^M A(1+j)^r = O(AM^{r+1})$, giving super-polynomially small total. Absorbing into the main-range bound yields (17) with $C(c, r) = 81 \Sigma(c, r)$. $\square$

Theorem 17 (Conditional-variance concentration). *As $M \rightarrow \infty$ in the joint regime,*

$$\text{Var} \left(\sum_{i=1}^k U_i \right) = O_c \left(\frac{1}{M^2 (\ln M)^4} \right) = o(s_k^4).$$

Consequently $\sum_i U_i / s_k^2 \xrightarrow{P} 1$.

Proof. We have

$$\text{Var} \left(\sum_i U_i \right) = \text{Var}(A) + 4 \text{Cov}(A, B) + 4 \text{Var}(B).$$

Set $J := M^{1/4}$. Terms involving a prime with $m_p > J$ will be called tail terms. Since

$$q_p^k \leq \exp \left(-\frac{km_p}{M-1} \right) \leq e^{-cJ/2}$$

for such primes and all sufficiently large M , every tail estimate below is super-polynomially small; polynomial factors in M and powers of $\log \log M$ are therefore harmless.

Bound on $\text{Var}(A)$. For $q_p \geq 1/2$,

$$c_{pp}^2 \text{Var}(R_{pp}^{(k)}) \leq \frac{\pi_p^2}{(p-1)^4} \frac{3q_p^k}{\pi_p^2} = \frac{3q_p^k}{(p-1)^4}.$$

The finitely many primes for which $q_p < 1/2$ contribute exponentially little. Lemma 16 therefore gives

$$\sum_p c_{pp}^2 \text{Var}(R_{pp}^{(k)}) = O_c\left(\frac{1}{M^3 \ln M}\right).$$

For the covariance terms with $m_p, m_q \leq J$, we have $pq > M$ and $\pi_p + \pi_q \leq 1/3$ for all sufficiently large M , so Lemma 15 applies. Hence

$$\sum_{\substack{p \neq q \\ m_p, m_q \leq J}} |c_{pp} c_{qq} \text{Cov}(R_{pp}^{(k)}, R_{qq}^{(k)})| \leq C_0(c)k \left(\sum_p \frac{\pi_p q_p^k}{(p-1)^2} \right)^2 = O_c\left(\frac{1}{M^3 (\ln M)^2}\right),$$

To justify the last estimate explicitly, set

$$S_M := \sum_{p \leq M} \frac{\pi_p q_p^k}{(p-1)^2}.$$

At level $m_p = j \leq M^{1/2}$, use $\pi_p = j/(M-1)$, $(p-1)^{-2} \leq 4(j+1)^2/M^2$, the count in Lemma 4, and $q_p^k \leq 2e^{-cj}$ to obtain

$$S_M \leq \frac{C}{M^2 \ln M} \sum_{j \geq 1} (j+1)^2 e^{-cj} + O(e^{-c\sqrt{M}/2}) = O_c\left(\frac{1}{M^2 \ln M}\right).$$

Since $k = O_c(M)$, this gives $kS_M^2 = O_c(M^{-3}(\ln M)^{-2})$ as claimed. Apart from the finitely many primes already absorbed above, if at least one of m_p, m_q exceeds J , Cauchy–Schwarz and Lemma 14 give

$$|c_{pp}| \sqrt{\text{Var}(R_{pp}^{(k)})} \leq \frac{\sqrt{3} q_p^{k/2}}{(p-1)^2},$$

and the resulting double sum is $O_c(e^{-cJ/4})$. Thus

$$\text{Var}(A) = O_c\left(\frac{1}{M^3 \ln M}\right). \quad (18)$$

A single-pair bound for $R_{pq}^{(k)}$. First suppose $m_p, m_q \leq J$. Then $pq > M$, so the events $\{p \mid X\}$ and $\{q \mid X\}$ are disjoint. Let $\beta = 1 - \pi_p - \pi_q$, $a = q_p q_q$, and $N = (T_p \wedge T_q) \wedge k$. Since

$$R_{pq}^{(k)} = \frac{a^k (a^{-N} - 1)}{1 - a},$$

we can bound its second moment directly. Put $u := \pi_p + \pi_q$ and $d := \pi_p \pi_q$, so that $\beta = 1 - u$ and $a = 1 - u + d$. Because $u \leq 1/3$ and $d \leq u^2/4$, one has $a^2 \leq \beta$ and $1 - a = u - d \geq u/2$. Therefore

$$\begin{aligned} a^{2k} \mathbb{E}[a^{-2N}] &= u \sum_{m=1}^k a^{2(k-m)} \beta^{m-1} + \beta^k \\ &\leq ku\beta^{k-1} + \beta^k \leq C(c)ku\beta^{k-1}, \end{aligned}$$

where the last step uses $ku \geq c/2$ for all sufficiently large M . Since $(a^{-N} - 1)^2 \leq a^{-2N}$,

$$\text{Var}(R_{pq}^{(k)}) \leq \mathbb{E}[(R_{pq}^{(k)})^2] \leq \frac{C(c)k\beta^{k-1}}{\pi_p + \pi_q}. \quad (19)$$

Using $|c_{pq}| \leq \pi_p \pi_q / ((p-1)(q-1))$, at a level pair (j_1, j_2) this gives

$$|c_{pq}| \sqrt{\text{Var}(R_{pq}^{(k)})} \leq \frac{C(c)}{M^3} \frac{(1+j_1)^2(1+j_2)^2}{\sqrt{j_1+j_2}} e^{-c(j_1+j_2)/2}. \quad (20)$$

Bound on $\text{Var}(B)$. Write $B = B_{\text{main}} + B_{\text{tail}}$, where B_{main} contains pairs with $m_p, m_q \leq J$. Minkowski's inequality in L^2 gives

$$\sqrt{\text{Var}(B_{\text{main}})} \leq \sum_{\substack{p < q \\ m_p, m_q \leq J}} |c_{pq}| \sqrt{\text{Var}(R_{pq}^{(k)})}.$$

By Lemma 4, the number of primes at level $j \leq J$ is at most $2.52M/(j \ln M)$. Combining this count with (20),

$$\sqrt{\text{Var}(B_{\text{main}})} \leq \frac{C(c)}{M(\ln M)^2} \sum_{j_1, j_2 \geq 1} \frac{(1+j_1)^2(1+j_2)^2}{j_1 j_2 \sqrt{j_1+j_2}} e^{-c(j_1+j_2)/2} = O_c\left(\frac{1}{M(\ln M)^2}\right),$$

because the double series converges.

It remains to control B_{tail} , which also includes every pair with $pq \leq M$. For arbitrary distinct p, q , let

$$\beta_{pq} := \mathbb{P}(p \nmid X, q \nmid X), \quad a := q_p q_q, \quad N := (T_p \wedge T_q) \wedge k.$$

If p is chosen so that $m_p = \max(m_p, m_q)$, then $a \leq q_p$ and $\beta_{pq} \leq q_p$. Directly from the geometric law of $T_p \wedge T_q$,

$$a^{2k} \mathbb{E}[a^{-2N}] = (1 - \beta_{pq}) \sum_{m=1}^k a^{2(k-m)} \beta_{pq}^{m-1} + \beta_{pq}^k \leq (k+1) q_p^{k-1}.$$

Since $1 - a \geq \pi_p$,

$$\text{Var}(R_{pq}^{(k)}) \leq \mathbb{E}[(R_{pq}^{(k)})^2] \leq \frac{(k+1) q_p^{k-1}}{\pi_p^2}. \quad (21)$$

For every tail pair, $m_p > J$ after relabelling. Using the crude bounds $|\kappa_{pq}| \leq 1$, $|\ln(1 - 1/r)| \leq 1/(r-1)$, $1/\pi_p \leq M$, and $\sum_{\substack{r \leq M \\ r \text{ prime}}} (r-1)^{-1} = O(\log \log M)$, Minkowski's inequality and (21) give

$$\sqrt{\text{Var}(B_{\text{tail}})} \leq CM^{3/2}(\log \log M)^2 e^{-cJ/4} = o(M^{-K})$$

for every fixed K . Therefore

$$\text{Var}(B) = O_c\left(\frac{1}{M^2(\ln M)^4}\right). \quad (22)$$

This argument controls all covariances among distinct prime-pair summands through Minkowski's inequality; no unproved orthogonality is used.

Conclusion. By Cauchy-Schwarz and (18)–(22),

$$|\text{Cov}(A, B)| \leq \sqrt{\text{Var}(A) \text{Var}(B)} = O_c\left(\frac{1}{M^{5/2}(\ln M)^{5/2}}\right) = o\left(\frac{1}{M^2(\ln M)^2}\right).$$

Thus

$$\text{Var}\left(\sum_i U_i\right) = O_c\left(\frac{1}{M^2(\ln M)^4}\right) = o(s_k^4),$$

because $s_k^4 \asymp M^{-2}(\ln M)^{-2}$ by Theorem 1. Finally, $\mathbb{E}[\sum_i U_i] = \sum_i \mathbb{E}[D_i^2] = s_k^2$, so Chebyshev's inequality gives $\sum_i U_i / s_k^2 \xrightarrow{P} 1$. $\square$

Lemma 18 (Linear-scale transfer). *Let $Y_k := \varphi(L_k)/L_k$ and $\mu_k := \mathbb{E} \ln Y_k$. In the joint regime,*

$$\frac{e^{\mu_k}}{P_M} \longrightarrow 1, \quad \frac{Y_k - \rho_k(M)}{P_M s_k} - \frac{W_k(M)}{s_k} \xrightarrow{P} 0.$$

Consequently the log-scale central limit theorem implies the linear-scale central limit theorem stated in Theorem 2.

Proof. Writing $a_p := \ln(1 - 1/p) < 0$,

$$\mu_k - \ln P_M = - \sum_{p \leq M} a_p q_p^k = \sum_{p \leq M} |a_p| q_p^k.$$

Partitioning by $m_p = j$, using $|a_p| \leq 1/(p-1)$ and Lemma 4 for $j \leq M^{1/2}$, gives

$$\sum_{p \leq M} |a_p| q_p^k \leq \frac{C(c)}{\ln M} \sum_{j \geq 1} e^{-cj} + o(1/\ln M) = O_c(1/\ln M).$$

The range $j > M^{1/2}$ is super-polynomially small. Hence $e^{\mu_k}/P_M = 1 + O_c(1/\ln M)$. By Mertens' product theorem [11], $P_M \asymp 1/\ln M$, so also $e^{-\mu_k} = O(\ln M)$.

Since $Y_k = e^{\mu_k + W_k} \leq 1$, we have $e^{W_k} \leq e^{-\mu_k}$. For $R_k := e^{W_k} - 1 - W_k$, Taylor's inequality gives

$$0 \leq R_k \leq \frac{1}{2} e^{(W_k)_+} W_k^2 \leq \frac{1}{2} e^{-\mu_k} W_k^2.$$

Therefore

$$\frac{\mathbb{E}|R_k|}{s_k} \leq \frac{1}{2} e^{-\mu_k} s_k = O\left(\sqrt{\frac{\ln M}{M}}\right) \longrightarrow 0,$$

where Theorem 1 was used. Since $\rho_k(M) = e^{\mu_k} \mathbb{E}[e^{W_k}]$,

$$\frac{Y_k - \rho_k(M)}{P_M s_k} = \frac{e^{\mu_k}}{P_M} \left(\frac{W_k}{s_k} + \frac{R_k - \mathbb{E}R_k}{s_k} \right).$$

The second term in parentheses tends to zero in L^1 , while W_k/s_k is tight because it has variance one. Together with $e^{\mu_k}/P_M \rightarrow 1$, this proves the claim. The same estimate also yields the Jensen-gap bound $0 \leq \rho_k(M) - e^{\mu_k} \leq s_k^2/2 = o(P_M s_k)$. $\square$

7 Proof of Theorem 2

For each M , define the row of the triangular array by $Z_{M,i} := D_i/s_k$, $1 \leq i \leq k = \lfloor cM \rfloor$, with row filtration $\mathcal{F}_{M,i} := \mathcal{F}_i$. Theorem 11 gives $\mathbb{E}[Z_{M,i} \mid \mathcal{F}_{M,i-1}] = 0$ and $\sum_i Z_{M,i} = W_k/s_k$. We use the Hall–Heyde martingale central limit theorem [5, Thm. 3.2] in the following standard form: it is enough that, for every $\varepsilon > 0$,

$$\sum_{i=1}^k \mathbb{E}[Z_{M,i}^2 \mathbf{1}\{|Z_{M,i}| > \varepsilon\} \mid \mathcal{F}_{M,i-1}] \xrightarrow{P} 0,$$

and that

$$\sum_{i=1}^k \mathbb{E}[Z_{M,i}^2 \mid \mathcal{F}_{M,i-1}] \xrightarrow{P} 1.$$

Corollary 13 verifies the first condition and Theorem 17 verifies the second. Hence $W_k/s_k \xrightarrow{d} \mathcal{N}(0, 1)$. Theorem 1 and Slutsky's theorem give $\sqrt{M \ln M} W_k / \sqrt{B(c)} \xrightarrow{d} \mathcal{N}(0, 1)$. Lemma 18 gives the corresponding linear-scale limit for $\varphi(L_k)/L_k - \rho_k(M)$. $\square$

8 Numerical Evidence

The theorems in Sections 3–7 are proved rigorously and require no numerical verification. The numerics below are included to illustrate finite- M behaviour, and for Theorem 2 to document that descriptive diagnostics of Gaussianity are consistent with the proof. The variance calculation evaluates the exact finite formula in extended-precision floating-point arithmetic, with an independent 80-decimal arbitrary-precision spot check. Monte Carlo sampling uses NumPy’s default PRNG, freshly initialised with the fixed seed 2026 for each panel.

Theorem 1. The rescaled variance $M \ln M \cdot s_k^2$ is evaluated deterministically from the exact finite formula $s_k^2 = V_1 + V_2$ via direct summation over $\mathcal{P}_M$ and $\mathcal{P}_M \times \mathcal{P}_M$. Figure 1 plots $M \ln M \cdot s_k^2$ versus M at $k = \lfloor cM \rfloor$ for $c \in \{0.5, 1, 2, 3\}$ and M up to 3200. Table 1 provides representative numerical values. Convergence to $B(c)$ is slow and visibly nonmonotone at finite M : at $M = 3200$, the rescaled variance agrees with $B(c)$ to within approximately 2% across $c \in [0.5, 3]$. The off-diagonal contribution V_2 is always negative (consistent with Remark 9) and an order of magnitude smaller than V_1 .

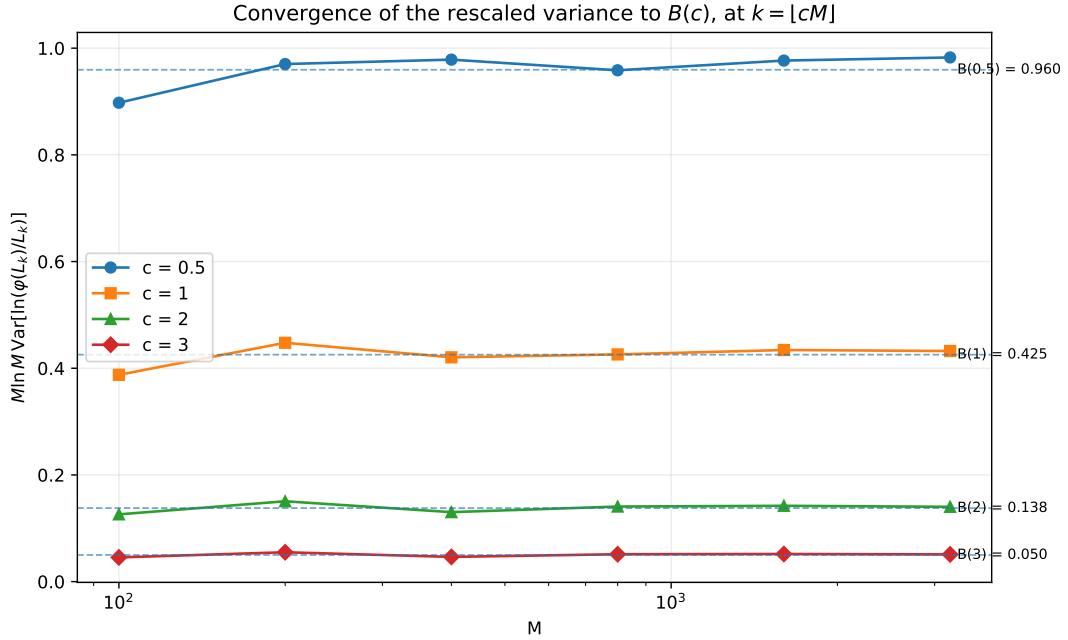


Figure 1: Rescaled variance $M \ln M \cdot s_k^2$ versus M at $k = \lfloor cM \rfloor$ for $c \in \{0.5, 1, 2, 3\}$. Values evaluated from the exact finite formula $s_k^2 = V_1 + V_2$ using extended-precision arithmetic. Dashed horizontal lines indicate the limit $B(c) = (e^c - 1)^{-1} - (e^{2c} - 1)^{-1}$ predicted by Theorem 1.

Theorem 2. Figure 2 displays the empirical distribution of the standardised coprime density $(\varphi(L_k)/L_k - \hat{\rho}_k)/\hat{\sigma}_k$ at $k = M$ for $M \in \{100, 300, 1000\}$, where $\hat{\rho}_k, \hat{\sigma}_k$ are the sample mean and sample standard deviation. Sample sizes are $N = 3000$ at $M = 100$ and $M = 300$; $N = 1000$ at $M = 1000$. Sample skewness decreases from 0.61 at $M = 100$ through 0.27 at $M = 300$ to 0.16 at $M = 1000$; excess kurtosis (Fisher convention) decreases from 0.58 at $M = 100$ through 0.12 at $M = 300$ to -0.18 at $M = 1000$. These are descriptive diagnostics rather than statistical tests; they are consistent with the proof above.

Table 1: Deterministic extended-precision evaluation of $M \ln M \cdot s_k^2$ at $k = \lfloor cM \rfloor$ across $c \in \{0.5, 1, 2, 3\}$ and $M \in \{100, 200, 400, 800, 1600, 3200\}$, alongside the limiting value $B(c)$ from Theorem 1. Values computed via $s_k^2 = V_1 + V_2$ by direct summation; no Monte Carlo.

M	$c = 0.5$	$c = 1.0$	$c = 2.0$	$c = 3.0$
100	0.8977	0.3877	0.1261	0.0454
200	0.9702	0.4478	0.1506	0.0550
400	0.9785	0.4205	0.1303	0.0464
800	0.9585	0.4259	0.1407	0.0514
1600	0.9767	0.4342	0.1422	0.0518
3200	0.9826	0.4323	0.1403	0.0510
$B(c)$	0.9595	0.4255	0.1379	0.0499

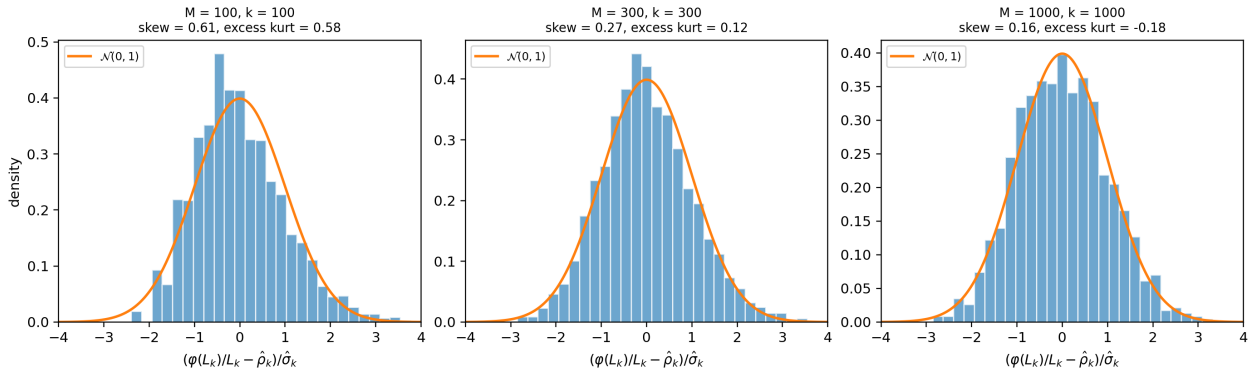


Figure 2: Empirical distribution of the standardised coprime density $(\varphi(L_k)/L_k - \hat{\rho}_k)/\hat{\sigma}_k$ at $k = M$, for three values of M . Solid curve: standard normal density. Sample sizes N and Fisher-convention skewness/excess kurtosis reported in panel titles. Histograms use 35 bins in $[-4, 4]$ and are normalised to density.

9 Discussion

Why the sup-norm bound needs both ingredients. The trivial Mertens-type bound $|D_i| \leq 2\Lambda(M) = O(\log \log M)$ is too large to feed into a Lyapunov condition: $(\log \log M)^2 \cdot M \cdot \ln M$ grows, not shrinks. A bound using only the coupon-collector estimate (restricting to primes with $m_p < (\log M)^2$, ignoring the sieve) yields $O(\log \log M)$ contributions per coordinate and is likewise insufficient. A bound using only the sieve-geometry (at most one contributing prime divides x) without the coupon-collector filter allows primes with very small q_p (i.e. very large m_p) that contribute as $O(1/\pi_p) = O(\log \log M)$ to the worst-case $|S_i(x)|$. Both estimates together reduce $|S_i(x)|$ to $(\log M)^2/(M-1)$, with the mean-correction $|C_i|$ contributing an additional $O((\log M)^3/M)$. The resulting $\|D_i\|_\infty = O((\log M)^3/M)$ is what the Hall–Heyde hypotheses need.

Finite- M convergence in Theorem 1. The proof establishes convergence but does not optimise a uniform remainder for the diagonal prime-level sum. The numerical values in Table 1 show slow, mildly oscillatory convergence at the tested values of M . A quantitative rate would require uniform error estimates in the partial-summation argument of Lemma 6.

General distributions. The Doob decomposition (Theorem 11) generalises verbatim to any distribution on $\{2, \dots, M\}$, with $\xi_p^{(i)}$ redefined as $\mathbf{1}\{p \mid X_i\} - \mathbb{P}(p \mid X)$. The sup-norm bound uses the uniformity of the X_i in two places: the coupon-collector estimate (which goes through if $\mathbb{P}(p \mid X) \gtrsim m_p/M$ uniformly) and the sieve-geometric observation (which is distribution-independent once levels are defined by q_p). The geometric disjointness in Proposition 5 is distribution-independent for any law supported on $\{2, \dots, M\}$: no support point is divisible by both p and q when $pq > M$. What is specific to the uniform law is the explicit formula $\pi_p = m_p/(M-1)$ and the resulting level structure; a non-uniform extension would require distribution-specific estimates for these probabilities and covariances. Extension to product distributions close to uniform appears feasible; substantial departures from uniformity would change the limit $B(c)$.

Comparison with fixed- k CLT results. Buraczewski–Iksanov–Marynych [3] established a CLT for $\log \text{lcm}$ of a uniform m -tuple on $\{1, \dots, n\}$ with m fixed as $n \rightarrow \infty$. Our result is not comparable in regime: they study $\log \text{lcm}$ itself, with k fixed, while we study $\log(\varphi(L_k)/L_k)$ with $k = \lfloor cM \rfloor$ and $M \rightarrow \infty$. The natural common ground — a CLT that interpolates between the two regimes, for instance $k = f(M)M$ with $f \rightarrow 0$ slowly — is not addressed here.

10 Conclusion

We have established, in the joint regime $k = \lfloor cM \rfloor$, $M \rightarrow \infty$, fixed $c > 0$, the variance asymptotic $M \ln M \cdot s_k^2 \rightarrow B(c) = (e^c - 1)^{-1} - (e^{2c} - 1)^{-1}$ (Theorem 1) and a central limit theorem for the standardised log coprime density (Theorem 2). The variance theorem is proved by a direct diagonal / off-diagonal decomposition, with the off-diagonal term controlled by an exact covariance identity (Lemma 7) valid for prime pairs (p, q) with $pq > M$. The CLT is proved via the Hall–Heyde martingale central limit theorem applied to the Doob filtration. Its two nontrivial ingredients are a uniform increment bound combining coupon-collector concentration with a sieve-geometric observation (Theorem 12), and a conditional-variance concentration resting on per-prime moment bounds, the covariance estimate of Lemma 15, and an L^2 Minkowski bound that controls all prime-pair cross terms. Together with the first-order results of the companion paper [12], these give a two-moment asymptotic theory for the coprime density in this regime.

Data and code availability. The numerical code, fixed random seed, generated figures, and machine-readable outputs underlying Table 1 and Figures 1–2 are available in the [public GitHub repository](#). The repository includes a one-command reproduction script and an arbitrary-precision validation mode for the exact variance formula.

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